

Model Performance Evaluation Strategy

I've researched similar model performances and have decided to use mean Average Precision (mAP) at Intersection over Union thresholds 0.5 and 0.5:0.95 (COCO-style). This means that if the predicted box overlaps the tagged box by 50% we count it as a correct prediction with the 0.5:0.95 being a sliding scale that penalizes 'loose' predictions. Using mAP allows me to figure out how well my model's recall *and* precision performs across all classifications, which in my project will be six defect classifications. To pair this back to the use case, I will frame recall and precision to what it actually means for a manufacturer shipping PCB boards. A false positive means that we would pull a working PCB board for repair when it doesn't need it and a false negative means a board with defects would be shipped to a customer. Both cost time and money, with the false negative likely costing more on both dimensions.

Deployment Strategy

For deployment, I plan to deploy the pipeline as a Docker container for the easiest possible deployment to different hardware. The container will house a Flask API and will use the YOLOv8 detector and ResNet classifier loaded in memory for inference. The API will accept images and return JSON with bounding box coordinates, classes and confidence scores. The reason I believe this is a solid solution for this use case is that in this industry I could imagine a manufacturer wanting to run this solution using both pre-saved data as well as in real-time with cameras. This would give this application the flexibility to be run whether on a cloud server processing batch uploads or an edge device on the factory floor connected to a camera

Optimization Plan

For optimization, I will take the YOLOv8 model and export it to ONNX. This will make the model runtime-agnostic. From there, I will apply ONNX runtime quantization to decrease the size and inference time for efficiency. I will make sure that I run benchmarking after the export to ONNX and after the quantization to ensure that recall and precision are not adversely affected (no more than 2% decrease in mAP and no more than 1% decrease in recall). This ties back to our initial manufacturing goals of avoiding false positives and false negatives. If performance is affected beyond the thresholds I have set, we will work on running inference with the original YOLOv8 model and explore other optimization options.